

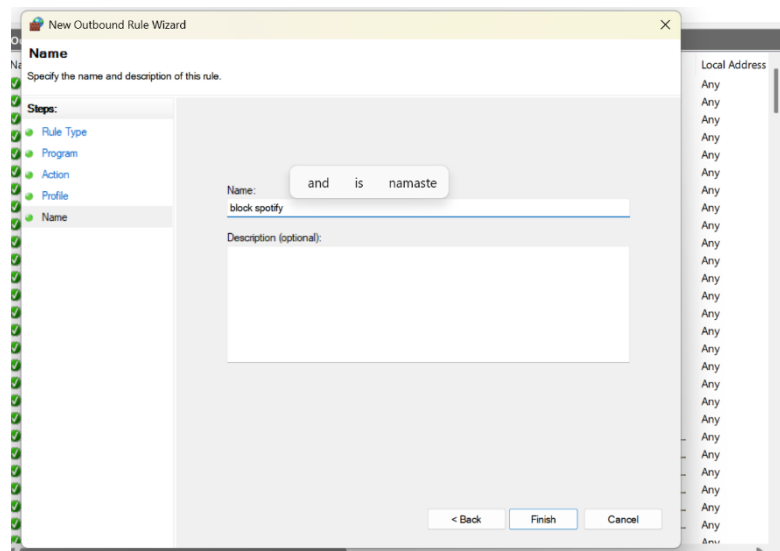
Day 4: Firewall BasicsTheory

Task 1 :- Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

ANS :-

- 1. Press Win + R, type wf.msc, and hit Enter.**
- 2. Click Outbound Rules on the left, then New Rule... on the far right.**
- 3. Keep Program selected and click Next.**
- 4. Paste this path into *This program path* and click Next:**
%APPDATA%\Spotify\Spotify.exe
- 5. Choose Block the connection, click Next, then click Next again.**
- 6. Name it Block Spotify and click Finish.**

*Practical Screenshot:-



Task 2 :- Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

ANS :-

1. Press Win + R, type wf.msc, and hit Enter.
2. Click Inbound Rules (left panel) → New Rule... (right panel).
3. Select Program → Click Next.
4. Paste this path into *This program path* → Click Next:
C:\Program Files\Google\Chrome\Application\chrome.exe
5. Select Allow the connection → Click Next → Click Next again.
6. Name it Allow Chrome 443 → Click Finish.
7. Double-click Allow Chrome 443 in the list → Go to Protocols and Ports tab.
8. Set Protocol type to TCP, Local port to Specific Ports, type 443 → Click OK.

Verification: Double-click the rule to confirm it shows TCP, port 443, and chrome.exe.

Task 3 :- Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

ANS :-

1. Click Inbound Rules (left panel) → New Rule... (right panel).
2. Select Custom → Click Next.
3. Leave All programs selected → Click Next → Click Next (Protocol).
4. Under *Which remote IP addresses does this rule apply to?*, select These IP addresses → Click Add...
5. Type 123.45.67.89 → Click OK → Click Next.
6. Select Block the connection → Click Next → Click Next.
7. Name it Block Bad IP → Click Finish.

Verification: The rule will appear in your list with a red block icon. Repeat under Outbound Rules if you want to block outgoing traffic to that IP as well.

Task 4 :- Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

ANS :-

3 Key Differences :-

- Where it runs : Hardware sits on a dedicated physical device at the network border. Software runs directly inside a computer's operating system.
- Scope : Hardware protects *all* devices on the network. Software protects *only* the single device it's installed on.
- Performance : Hardware processes traffic using its own chips without slowing down PCs. Software uses your computer's CPU and RAM.

Use Cases :-

- Hardware Firewall : Large Office — Blocks external threats at the front door before malicious traffic ever reaches company computers, Wi-Fi, or servers.
- Software Firewall : Remote / Home Worker — Protects a laptop when connecting to untrusted public Wi-Fi networks (like at a coffee shop) where a physical network firewall isn't present.